IMovies – Movie Ticket Booking System

Requirement Analysis

Software requirements specification

Project documentation based on the uploaded bookMyShow-main source ZIP

Prepared: 31 October 2026

Requirement Analysis

2.1 Purpose

This SRS defines the functional and quality requirements for the IMovies system represented by the supplied source archive.

2.2 User classes

Role	Responsibilities
Visitor	Browse public movie listings and register or log in.
Registered user	Access account features, choose shows/seats and review bookings.
Administrator / theatre manager	Manage movies, theatres and show schedules through protected interfaces.

2.3 Functional requirements

- FR-01: The system shall allow a visitor to create an account using name, email and password.
- FR-02: The system shall authenticate a registered user and issue a JWT token.
- FR-03: The system shall display the movie catalogue and movie details.
- FR-04: The system shall display theatres and shows for a selected movie.
- FR-05: The system shall display seat availability and allow seat selection, subject to the six-seat limit described in the README.
- FR-06: The system shall submit a booking and update the selected show's booked-seat list.
- FR-07: The system shall allow an authenticated user to view booking records.
- FR-08: Authorized administrators shall be able to add, update and delete movie records.
- FR-09: Authorized administrators shall be able to manage theatres and show schedules.

2.4 Non-functional requirements

- Security: passwords should be stored as hashes; protected API routes should validate JWTs.
- Usability: navigation and booking steps should be understandable to a first-time user.
- Performance: common catalogue and show requests should return within an acceptable response time under the intended classroom demo load; measure rather than assume.
- Reliability: booking operations should avoid assigning the same seat to multiple customers; verify concurrency handling before claiming this guarantee.
- Maintainability: frontend pages, API route files, database models and middleware are separated into modules.
- Compatibility: target a modern desktop browser and a supported Node.js runtime.

2.5 Software requirements

Component	Technology evidenced in package files
Frontend	React 18, React Router, Redux Toolkit, Axios, Ant Design, Moment
Backend	Node.js, Express 4
Database	MongoDB with Mongoose 8
Authentication	jsonwebtoken and bcryptjs

Component	Technology evidenced in package files
Other dependency	Stripe SDK is present in server dependencies; payment flow/configuration must be verified.

2.6 Hardware requirements

Development/demo baseline: laptop or desktop with a modern dual-core CPU, 8 GB RAM recommended, internet connection for Atlas/remote assets, and a current Chrome/Firefox/Edge browser. These are practical recommendations, not measured minimums.

2.7 Constraints and assumptions

The README recommends Node.js 16+ and MongoDB local or Atlas. The backend listens on port 8082. The frontend uses Create React App scripts (react-scripts start), despite a README line referring to Vite. The actual package.json is the authoritative setup reference.

2.8 Acceptance criteria

A successful demo should show account registration/login, movie browsing, theatre/show discovery, seat selection, booking confirmation and booking-history retrieval, plus administrator management if an admin account is available.

Accuracy note: This document describes the code and README found in the supplied ZIP. Items marked as recommendations or proposed tests are not claims that they have already been implemented or passed. Confirm all workflows on your laptop before submission.